MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 01** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer question (2 marks) and 3 Short Answer questions (3 marks each).
4. **Section C** comprises 1 Applied Case Study question of 4 marks with internal choice.
5. Use of calculators is not permitted.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. In an AP, if $d = -4$, $n = 7$, and $a_n = 4$, then the first term a is: [1]

(A) 6 (B) 7 (C) 20 (D) 28

Q2. In an AP, if $a = 3.5$, $d = 0$, and $n = 101$, then a_n equals: [1]

(A) 0 (B) 3.5 (C) 103.5 (D) 104.5

Q3. The list of numbers $-10, -6, -2, 2, \ldots$ is: [1]

(A) an AP with $d = -16$ (B) an AP with $d = 4$ (C) an AP with $d = -4$ (D) not an AP

Q4. The 11th term of the AP $-3, -\frac{1}{2}, 2, \ldots$ is: [1]

(A) 28 (B) 22 (C) -38 (D) $-46\frac{1}{2}$

Q5. **Assertion (A):** The sequence $-10, -6, -2, 2, \ldots$ forms an AP with common difference 4.
Reason (R): A sequence $a_1, a_2, a_3, \ldots$ is an AP if $a_{k+1} - a_k$ is constant for all $k \geq 1$. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Short Answer Questions (11 Marks)

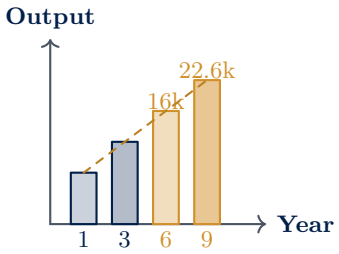
Q6. Find the 20th term of the AP $4, 9, 14, 19, \ldots$. [2]

Q7. The sum of the 4th and 8th terms of an AP is 24 and the sum of the 6th and 10th terms is 44. Find the first three terms of the AP. [3]

Q8. The pth term of an AP is q and its qth term is p. Prove that its nth term is $(p + q - n)$, and deduce that its $(p + q)^{\text{th}}$ term is zero. [3]

Q9. If the ratio of the sums of first n terms of two APs is $(7n+1) : (4n+27)$, find the ratio of their 11th terms. [3]

Q10. CASE STUDY: Manufacturing Production Schedule (Smart TV Factory)



A consumer electronics manufacturing plant in Greater Noida produces high-definition smart LED television sets. To meet rising market demand, management modernizes the plant with automated robotic assembly lines.
Consequently, the annual production of TV sets increases uniformly by a constant number of units every year. The factory produced $16,000$ sets in its 6th year of commercial operation and $22,600$ sets in its 9th year.

Based on the above mathematical scenario, answer the following questions:

(i) Find the fixed annual increase in production (d). **[1]**

(ii) Determine the production of smart TV sets during the first year of operation (a). **[1]**

(iii) Calculate the total cumulative production of smart TV sets manufactured in the first 8 years. **[2]**

— **OR** —

In which operational year will the annual production reach $29,200$ smart TV sets? **[2]**

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